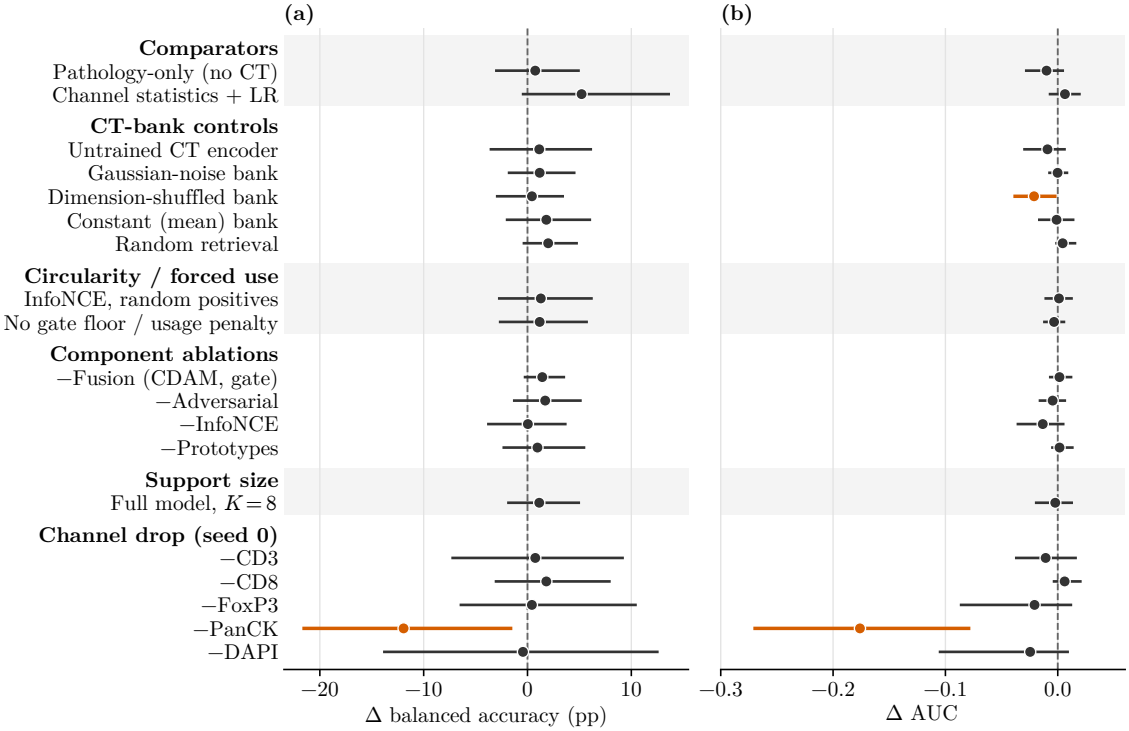
(a)
(b)
Comparators
Pathology-only (no CT)
Channel statistics + LR
CT-bank controls
Untrained CT encoder
Gaussian-noise bank
Dimension-shuffled bank
Constant (mean) bank
Random retrieval
Circularity / forced use
InfoNCE, random positives
No gate floor / usage penalty
Component ablations
−Fusion (CDAM, gate)
−Adversarial
−InfoNCE
−Prototypes
Support size
Full model, $K = 8$
Channel drop (seed 0)
−CD3
−CD8
−FoxP3
−PanCK
−DAPI
−20
−10
0
10
Δ balanced accuracy (pp)
−0.3
−0.2
−0.1
0.0
Δ AUC